EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The **Bilinear Transformation (BLT)** is the most widely used algebraic method for designing IIR digital filters from analog prototypes. By applying the trapezoidal rule of numerical integration, the entire infinite continuous imaginary axis $j\Omega$ is mapped monotonically onto the finite unit circle $e^{j\omega}$, **completely eliminating spectral aliasing**.

Pedagogical Strategy:

1. Derive the Bilinear mapping from trapezoidal numerical integration of $\frac{dy}{dt} = x(t)$.
2. Establish the conformal mapping relation: $s = \frac{2}{T_d} \frac{1-z^{-1}}{1+z^{-1}}$.
3. Prove that the left-half s-plane maps strictly inside the unit circle $|z| < 1$ (guaranteed stability).
4. Derive the **Non-linear Frequency Warping relationship**:
$$\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right) \iff \omega = 2\arctan\left(\frac{\Omega T_d}{2}\right)$$
5. Master **Frequency Prewarping**: Prewarping digital critical frequencies $\omega_p, \omega_s \to \Omega_p, \Omega_s$ before designing the analog prototype.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Bilinear Transformation algebraic mapping and frequency warping relation.
2. **Prewarp** digital filter specifications into continuous analog frequencies.
3. **Design** Butterworth and Chebyshev IIR digital filters using the Bilinear Transformation.
4. **Compare** the Bilinear Transformation with the Impulse Invariance method.

2. MATHEMATICAL FOUNDATIONS

2.1 Derivation via Trapezoidal Integration

Consider the first-order differential equation $\frac{dy(t)}{dt} = x(t)$. Integrating over $[(n-1)T_d, nT_d]$ using the trapezoidal rule:

$$y(nT_d) - y((n-1)T_d) = \int_{(n-1)T_d}^{nT_d} x(t)dt \approx \frac{T_d}{2}[x(nT_d) + x((n-1)T_d)]$$

In discrete notation:

$$y[n] - y[n-1] = \frac{T_d}{2}[x[n] + x[n-1]]$$

Taking the Z-Transform:

$$Y(z)(1 - z^{-1}) = \frac{T_d}{2} X(z)(1 + z^{-1}) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{T_d}{2} \frac{1 + z^{-1}}{1 - z^{-1}}$$

Since the analog integrator transfer function is $H_a(s) = \frac{1}{s}$, equating $H_a(s) = H(z)$ yields:

$$\mathbf{s = \frac{2}{T_d} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{2}{T_d} \frac{z - 1}{z + 1}}$$

2.2 Frequency Warping and Prewarping

Substitute $s = j\Omega$ and $z = e^{j\omega}$:

$$j\Omega = \frac{2}{T_d} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T_d} \frac{e^{j\omega/2} - e^{-j\omega/2}}{e^{j\omega/2} + e^{-j\omega/2}} = j \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

$$\mathbf{\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)}$$

- **Prewarping Formulas:**

$$\Omega_p = \frac{2}{T_d} \tan\left(\frac{\omega_p}{2}\right), \qquad \Omega_s = \frac{2}{T_d} \tan\left(\frac{\omega_s}{2}\right)$$

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3. WORKED NUMERICAL EXAMPLES

Example 28.1: 1st-Order Lowpass Filter via BLT

Problem: Design a digital lowpass filter with a 3-dB cutoff frequency $\omega_c = 0.2\pi \text{ rad/sample}$ from an analog prototype $H_a(s) = \frac{\Omega_c}{s + \Omega_c}$ using the Bilinear Transformation ($T_d = 1 \text{ s}$).

Solution:

1. **Prewarp the Cutoff Frequency:**

$$\Omega_c = \frac{2}{T_d} \tan\left(\frac{\omega_c}{2}\right) = \frac{2}{1} \tan(0.1\pi) = 2 \times 0.3249 = 0.6498 \text{ rad/s}$$

2. **Analog Transfer Function:**

$$H_a(s) = \frac{0.6498}{s + 0.6498}$$

3. **Apply Bilinear Transformation ($s = 2\frac{1 - z^{-1}}{1 + z^{-1}}$):**

$$H(z) = \frac{0.6498}{s + 0.6498}\Bigg|_{s = 2\frac{1 - z^{-1}}{1 + z^{-1}}} = \frac{0.6498}{2\frac{1 - z^{-1}}{1 + z^{-1}} + 0.6498} = \frac{0.6498(1 + z^{-1})}{2(1 - z^{-1}) + 0.6498(1 + z^{-1})}$$

$$H(z) = \frac{0.6498 + 0.6498z^{-1}}{(2 + 0.6498) + (-2 + 0.6498)z^{-1}} = \frac{0.6498(1 + z^{-1})}{2.6498 - 1.3502z^{-1}} = \frac{0.2452(1 + z^{-1})}{1 - 0.5095z^{-1}}$$

4. **Verification:**
   - At DC ($\omega = 0, z = 1$): $H(1) = \frac{0.2452(2)}{1 - 0.5095} = \frac{0.4904}{0.4905} \approx 1.0$ (0 dB).
   - At half-sampling ($\omega = \pi, z = -1$): $H(-1) = \frac{0.2452(0)}{1 + 0.5095} = 0$ ($-\infty$ dB).

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Bilinear Transformation and the frequency warping equation. Explain the necessity of prewarping. *(8 Marks)* **(b)** Design a digital Butterworth lowpass filter satisfying:

- $0.8 \le |H(e^{j\omega})| \le 1.0$ for $0 \le \omega \le 0.2\pi$
- $|H(e^{j\omega})| \le 0.2$ for $0.6\pi \le \omega \le \pi$ Use Bilinear Transformation with $T_d = 1$ s. *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Trapezoidal derivation of $s = \frac{2}{T_d}\frac{1-z^{-1}}{1+z^{-1}}$ *(4 Marks)*
  - Warping derivation $\Omega = \frac{2}{T_d}\tan(\omega/2)$ and prewarping explanation *(4 Marks)*
- **Part (b):**
  - Prewarp: $\Omega_p = 2\tan(0.1\pi) = 0.6498, ; \Omega_s = 2\tan(0.3\pi) = 2.7528$ *(2 Marks)*
  - Specs: $A_p = -20\log_{10}(0.8) = 1.9382 \text{ dB}, ; A_s = -20\log_{10}(0.2) = 13.9794 \text{ dB}$ *(1 Mark)*
  - Order calculation: $N \ge \frac{\log_{10}[(10^{1.398}-1)/(10^{0.1938}-1)]}{2\log_{10}(2.7528/0.6498)} = \frac{\log_{10}(24/0.5623)}{2\log_{10}(4.236)} = \frac{1.6303}{1.254} = 1.30 \implies \mathbf{N=2}$ *(2 Marks)*
  - Synthesize 2nd-order Butterworth $H_a(s)$ and substitute $s \to 2\frac{1-z^{-1}}{1+z^{-1}}$ *(2 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

# Butterworth BLT design
b, a = signal.butter(2, 0.2, btype='low', analog=False)
print("Digital Filter b:", np.round(b, 4))
print("Digital Filter a:", np.round(a, 4))
```